Boehringer Ingelheim

Impurities

Tenecteplase	Global ID_Version 206893_11135124_1.0
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ACRONYMS AND ABBREVIATIONS

CGE	Capillary gel electrophoresis
CHO	Chinese hamster ovary
DF	Diafiltration
DNA	Desoxyribonucleic acid
DS	Drug substance
DP	Drug product
ELAC	Epoxy lysine agarose chromatogel
ELISA	Enzyme-linked immunosorbent assay
ha-PA	Hamster plasminogen activator
HCP	Host cell proteins
HIC	Hydrophobic interaction chromatography
HMW	High molecular weight
HPLC-FLD	High performance liquid chromatography with fluorescence detection
HPSEC	High performance size exclusion chromatography
ICP-OES	Inductively coupled plasma optical emission spectrometry
LC-ESI-MS/MS	Liquid chromatography / electrospray ionization-tandem mass spectrometry
LMW	Low molecular weight
LOD	Limit of detection
LOQ	Limit of quantitation
MTX	Methotrexate
PDE	Permissible daily exposure
PDMS	Polydimethylsiloxane
pPRI	Potential process-related impurity
PRI	Process-related impurity
PPQ	Process performance qualification
qPCR	Quantitative polymerase chain reaction
RP-HPLC	Reversed phase high performance liquid chromatography
SDS-PAGE	Sodium dodecyl sulphate polyacrylamide gel electrophoresis
Tenecteplase	Recombinant human tissue-type plasminogen activator
TTC	Threshold of toxicological concern
UF	Ultrafiltration
WHO	World Health Organization

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1. INTRODUCTION

This section summarises the impurities that are controlled and monitored during the tenecteplase drug substance (DS) manufacturing process.

Impurities classified as either process-related or product-related, are monitored during purification as a measure of process robustness and to ensure patient safety. The tenecteplase DS purification process was developed to remove product- and process-related impurities.

Process-related impurities (PRIs) are defined as impurities that are derived from the manufacturing process and arise from substances introduced during cell culture or purification, or may be produced by the host cells, Chinese hamster ovary (CHO) cells. A risk assessment based on scientific understanding of the purification process described in Chapter 2, was used for identification, categorisation and measurement of potential PRIs (pPRIs).

For tenecteplase, the consistent and efficient removal of PRIs during DS manufacture, including host cell deoxyribonucleic acid (DNA), host cell-derived proteins (HCP), hamster plasminogen activator (ha-PA), insulin, poloxamer 188, propylene glycol and gentamicin was demonstrated by in-process testing and / or DS release testing during process performance qualification (PPQ); see Sections 3.2.S.2.5 Process Validation and / or Evaluation and 3.2.S.2.6 Manufacturing Process Development). In addition, host cell DNA, insulin, poloxamer 188, gentamicin, phenol red, methotrexate (MTX) and simethicone were analysed with retains of in-process and / or DS samples from already released commercial manufacturing batches.

For the adapted Phenyl Sepharose FF hydrophobic interaction chromatography (HIC) step (for details see Section 3.2.S.2.6 Manufacturing Process Development), the third and last chromatography step in the purification process, the removal of the PRIs HCP, DNA, and ha-PA to levels below limit of quantitation (LOQ) or to negligible concentrations was demonstrated based on the data obtained for three consecutive manufacturing runs during process revalidation (12650000 – 12650002; see Sections 3.2.S.2.5 Process Validation and / or Evaluation, 3.2.S.2.6 Manufacturing Process Development). Additionally, a small-scale DNA spiking study confirmed that the DNA removal capacity of the adapted HIC remained comparable to the initial process. No additional testing for insulin, poloxamer 188, gentamicin, MTX, phenol red or simethicone was conducted. Levels for these impurities are already below or close to the limit of detection (LOD) or LOQ after the first chromatography step. The gentamicin spiking study confirmed the high removal capability of the first two chromatography steps and the final ultrafiltration / diafiltration (UF / DF). Furthermore, no additional testing for propylene glycol was performed because its amount and the time point of addition during the adapted HIC are identical to those in the initial process.

Based on the analytical results presented in Chapter 2 and in Sections 3.2.S.2.5 Process Validation and / or Evaluation and 3.2.S.2.6 Manufacturing Process Development, the consistent and efficient removal of the evaluated PRIs was demonstrated.

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Product-related impurities are molecular variants of the active molecule that can arise during the manufacturing process of tenecteplase DS or drug product (DP), which do not have properties comparable to those of the desired product with respect to efficacy and safety. Identified product-related impurities are the high molecular weight (HMW) and low molecular weight (LMW) species. A detailed description of these molecular variants and their occurrence is provided in Section 3.2.S.3.1 Elucidation of Structure and other Characteristics. Data obtained for product-related impurities demonstrating the efficient control by the purification process are presented in Chapter 3.

No new product-related impurities are introduced during the manufacture of tenecteplase DP (refer to Section 3.2.P.5.5 Impurities).

2. PROCESS-RELATED IMPURITIES

The clearance assessment for PRIs followed a risk-based approach to assess the potential impact and clearance of PRIs that are not routinely monitored for the quality of the product. For the assessment all soluble raw materials of the cell culture and purification process were assumed to be potential PRIs (pPRI).

The theoretical clearance calculation was performed using worst-case assumptions in a fate assessment. For the fate assessment, the maximum amount of raw materials entering the process, the minimum DS volume and the maximum human dose of 50 mg per day were applied as worst-case assumptions to calculate the theoretical dose of PRI per dose. The contribution of the 10 kDa UF / DF and chromatography steps for clearance of a particular PRI was based on its properties (e.g., molecular weight, potential for micelle formation, charge) and the point of addition in the process.

For the safety assessment pPRIs from the tenecteplase DS manufacturing process were assigned toxicological thresholds based on categorical thresholds of toxicological concern (TTC) or PRI-specific determination of permissible daily exposure (PDE) levels.

The TTC is defined as the level that represents a negligible risk of toxicity or carcinogenicity and was set based on toxicological potential:

- Any pPRI reported to be genotoxic was assigned a TTC of 1.5 µg/day
- Any pPRI considered to be an irritant or sensitising agent was assigned a TTC of 5 µg/day

PDE values were based on available toxicology information and considered the route of administration and application of adequate safety factors.

Analytical measurements were performed for all pPRIs for which the calculated levels exceeded the daily toxicological threshold (insulin, poloxamer 188). Furthermore, the clearance of the following pPRIs was verified for the following reasons although their calculated levels were below the respective toxicological threshold:

- MTX and phenol red due their genotoxic potential

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- gentamicin due to its potential as irritant and pharmacological relevance (antibiotic)
- simethicone due to its pharmacological relevance
- propylene glycol to demonstrate, by way of example, the consistent removal of a substance which is introduced during the last chromatography step preceding the final UF / DF

The host cell-derived impurities DNA, HCP and ha-PA were measured via testing of in-process and / or DS samples to provide confirmation of depletion of these pPRIs.

A summary of identified PRIs for evaluation, their sources and the corresponding toxicological threshold are listed in Table 1. The consistent removal of these PRIs is demonstrated below in Chapters 2.1 to 2.10.

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Table 1 Process-related impurities

Process-related impurity	Source	Amount in single dose after theoretical fate assessment *	Toxicological threshold	Method of control
DNA	Host cell	n/a	World Health Organization (WHO) limit of ≤ 10 ng/dose	In-process testing during PPQ and in selected commercial manufacturing batches; spiking study
HCP	Host cell	n/a	DS release specification < 500 ppm	In-process testing during PPQ and in selected commercial manufacturing batches; DS specification
ha-PA	Host cell	n/a	Maximum patient exposure < 38,750 ng per dose #	In-process testing during PPQ and in selected commercial manufacturing batches
Insulin	Cell culture media	797 µg/dose	PDE of 1.0 µg/day	In-process testing during PPQ and in selected commercial manufacturing batches
Poloxamer 188	Cell culture media	81,126 µg/dose	PDE of 16,500 µg/day	In-process testing during PPQ and in selected commercial manufacturing batches
Gentamicin	Cell culture media	0.17 ng/dose	TTC of 5.0 µg/day	In-process testing during PPQ and in selected commercial manufacturing batches; spiking study
MTX	Additive during inoculum train	0.008 pg/dose	TTC of 1.5 µg/day	In-process testing during PPQ and in selected commercial manufacturing batches
Phenol red	Cell culture media	0.046 pg/dose	TTC of 1.5 µg/day	In-process testing in selected commercial manufacturing batches
Simethicone	Additive during cell culture manufacturing process	1,065 µg/dose	PDE of 2,500 µg/day	In-process testing in selected commercial manufacturing batches
Propylene glycol	Buffer component during purification manufacturing process	196 µg/dose	PDE of 125,000 µg/day	In-process testing during PPQ

* calculated based on the 12,000 L process with a maximum daily dose of 50 mg

\# 775 ppm = 775 ng/mg, 50 mg dose

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2.1 DNA

Tenecteplase is produced in CHO cells, thus residual DNA from the cell line may be present in the DS. The tenecteplase purification process, especially the unit operations epoxy lysine agarose chromatogel (ELAC) affinity chromatography and HIC, was developed for DNA removal. In compliance with the WHO guideline, no more than 10 ng of DNA per dose are allowed. For a maximum dose of 50 mg, this correlates to a maximum allowable DNA concentration of 200 pg/mg tenecteplase.

Applying a radioactive DNA hybridisation assay, the initial process validation studies demonstrated that the commercial manufacturing process is capable to reproducibly remove DNA to levels below the LOD (DS batches 22000 – 22002, refer to Section 3.2.S.2.5 Process Validation and / or Evaluation). The maximum theoretically calculated level of no more than 20 pg DNA/mg tenecteplase is at least 10-fold below the WHO limit.

Furthermore, in a spiking study with radiolabelled DNA at small scale, the removal capacity of the initial HIC and the overall DNA binding capacity of the HIC resin was assessed. A DNA removal of at least 3.46 $\log_{10}$ with DNA loads in the range of 146 ng DNA to 375 µg DNA per mL resin was demonstrated.

In addition, analytical testing of retains of in-process and DS samples from already released commercial manufacturing batches by quantitative polymerase chain reaction (qPCR) was performed. As shown in Table 2, DNA removal during purification to levels close to or below LOQ in DS was found to be consistent, reproducible, and highly efficient for all batches. The main DNA removal steps are the ELAC affinity chromatography and the HIC. The latter step follows the cation exchange chromatography. All measured values in the DS are below the toxicological threshold. The theoretical maximum DNA application per daily dose of 50 mg tenecteplase DS would be 70 pg.

For the adapted HIC, a small-scale spiking study with a DNA spike of 1,000,000 pg per mg tenecteplase demonstrated a DNA removal factor of at least 8,014 (3.90 $\log_{10}$). This study confirmed that the adaptations in the HIC step have no impact on the overall DNA removal capability of this step and that the demonstrated removal factor is comparable to the former DNA clearance study.

Furthermore, the DNA content measured by qPCR for the respective revalidation runs 12650000 – 12650002 of the process with the adapted HIC is listed in Table 3. All values for the HIC and the DS are below the LOQ. Based on the DNA content in the DS of < 6 pg DNA/mg protein, this results in a theoretical DNA content below the toxicological threshold of < 300 pg DNA per 50 mg dose. The differences in the theoretically calculated values below the LOQ in Table 2 and Table 3 result only from a different pre-dilution of the samples prior to DNA extraction.

Since the removal capability of the tenecteplase purification process regarding DNA has been shown to be efficient and reproducible for all batches investigated, monitoring was discontinued.

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Table 2 DNA removal during purification

DS batch	Amount of DNA [pg/mg]			
	Harvested cell culture fluid	ELAC affinity chromatography	Cation exchange chromatography	DS
12630004	74,979,484	449	134	< 1.20 [a]
12630005	89,788,308	416	162	1.39
12630013	235,539,572	1615	374	< 1.20 [a]
12630014	74,571,808	1049	480	< 1.20 [a]
12630020	97,793,799	807	182	< 1.20 [a]
12630022	114,784,397	755	163	< 1.19 [a]
12630034	211,266,804	2007	345	< 1.19 [a]

a value below LOQ, thus value calculated theoretically considering LOQ, sample dilution and protein concentration

Table 3 Residual DNA results for the adapted HIC revalidation runs

DS batch	Amount of DNA [pg/mg]	
	Adapted HIC	DS
12650000	< 30.0 [a]	< 5.8 [a]
12650001	< 29.9 [a]	< 5.9 [a]
12650002	< 29.4 [a]	< 5.9 [a]

a value below LOQ, thus value calculated theoretically considering LOQ, sample dilution and protein concentration

Analytical procedure
Residual host cell DNA was measured via a CHO specific quantitative polymerase chain reaction (qPCR) DNA. The method consists of two steps: the DNA extraction and quantification by qPCR.
During the purification of nucleic acids, extracted DNA is bound to small metal particles (magnetic beads) coated with a silica matrix DNA binding substance. The samples are then exposed to a strong magnetic field which binds the particles to a magnetic probe. After removing the liquid and several wash steps, the DNA is eluted from the beads with an appropriate buffer.

The real-time PCR (TaqMan format) quantifies DNA over the entire course of amplification. Samples are incubated with DNA polymerase, deoxyribonucleotide triphosphates, primers specific for CHO DNA, and a TaqMan reporter probe. The probe is a small CHO DNA specific oligonucleotide with a fluorescent reporter dye on the 5´end and a quencher dye on the 3´end. When intact, the proximity of the quencher dye greatly reduces the fluorescence dye by fluorescence resonance energy transfer (FRET). If the target is present, the probe anneals downstream of the primer and is cleaved by the 5´nuclease activity of the Taq DNA

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polymerase as the primer is extended. This cleavage separates the reporter dye from the quencher dye, increasing the reporter fluorescence and removes the reporter strand sot that the extension of the strand continues to the end. This fluorescence signal is directly proportional to the concentration of CHO DNA in the sample. The range of the exponential increase in the product is used to determine the amount of DNA in the sample. Using a calibration curve generated by samples containing known amounts of CHO DNA, unknown samples can be quantified by regression analysis.

The LOQ of the analytical method is 0.03 pg DNA per PCR performed with a sample volume of 10 µL.

Summary
It is demonstrated that the tenecteplase DS manufacturing process is capable of reproducibly removing DNA to levels below the DS release criterion of no more than 200 pg/mg protein. DNA levels in DS are at least 33-fold below the WHO limit of 10 ng DNA for a maximum dose of 50 mg tenecteplase.

2.2 HCP

Tenecteplase is produced in CHO cells, thus residual proteins from the cell line may be present in the DS. In the purification process, the chromatography steps were developed for HCP removal. Major reduction of HCP occurs at the ELAC affinity chromatography step. The subsequent chromatography steps contribute to further clearance of HCP.

The initial process validation studies (DS batches 22000 – 22002, refer to Sections 3.2.S.2.5 Process Validation and / or Evaluation and 3.2.S.2.6 Manufacturing Process Development) demonstrated that the removal of HCP during purification is consistent, reproducible, and sufficient. This is supported by the analytical testing of retains of in-process samples from already released commercial manufacturing batches (Table 4) and the analytical results for the adapted HIC revalidation runs 12650000 – 12650002 listed in Table 5.

Since the removal capability of the tenecteplase purification process regarding HCP has been shown to be efficient and reproducible for all batches investigated, monitoring on in-process samples was discontinued. The level of HCP is controlled as part of the DS release testing.

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Table 4 HCP removal during purification

DS batch	Amount of HCP [ppm]				
	Harvested cell culture fluid	ELAC affinity chromatography	Cation exchange chromatography	HIC	DS
12630004	617,987	302	194	75	105
12630005	597,108	223	131	78	89
12630013	552,999	415	239	113	163
12630014	273,803	474	202	109	162
12630020	630,305	371	174	116	97
12630022	582,221	505	239	103	108
12630034	470,644	669	331	130	136

Table 5 HCP results for the adapted HIC revalidation runs

DS batch	Amount of HCP [ppm]			
	ELAC affinity chromatography	Cation exchange chromatography	Adapted HIC	DS
12650000	364	< 175 [a]	73	83
12650001	422	< 175 [a]	81	78
12650002	255	< 175 [a]	63	< 75 [a]

a value below LOQ, thus value calculated theoretically considering LOQ, sample dilution and protein concentration

Analytical procedure
The analytical procedure used to measure HCP in retains of commercial manufacturing and revalidation batches is the same assay as used for release testing of tenecteplase DS. For a description of the CHO protein-specific enzyme-linked immunosorbent assay (ELISA), please refer to Section 3.2.S.4.2 Analytical Procedure.

Summary
The HCP removal capacity of the tenecteplase purification process was sufficiently shown for materials from the manufacturing scale. The purification process reproducibly reduces HCP to levels below the DS release criterion of less than 500 ppm in tenecteplase DS.
The LOQ of the analytical method is 6.25 ng/mL.

2.3 HAMSTER PLASMINOGEN ACTIVATOR

Tenecteplase is produced in CHO cells, thus residual ha-PA from the cell line may be present in the DS.

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The initial process validation studies (DS batches 22000 – 22002, refer to Section 3.2.S.2.5 Process Validation and / or Evaluation) demonstrated an overall 3-fold removal of ha-PA by the ELAC affinity chromatography and HIC steps.

The consistent removal of ha-PA is supported by the comparability results shown in Section 3.2.S.2.6 Manufacturing Process Development and the analytical testing of in-process and DS samples from commercial manufacturing batches shown in Table 6 and the analytical results for the adapted HIC revalidation runs 12650000 – 12650002 listed in Table 7. Compared to the commercial batches shown in Table 6, for the adapted HIC the ha-PA level after HIC and DS is higher because, due to the increased ammonium sulfate content, ha-PA shows increased interaction to the HIC resin and is co-purified with tenecteplase. However, the ha-PA levels after HIC are still lower than in the initial process validation runs (324 - 404 ppm).

The purification capability of the tenecteplase purification process regarding ha-PA has been shown to be efficient. Levels of ha-PA were reproducible below the toxicological threshold of 38,750 ng per maximum dose of 50 mg for all batches investigated. Therefore, monitoring was discontinued.

Table 6 ha-PA removal during purification

DS batch	Amount of ha-PA [ppm]		
	Harvested cell culture fluid	HIC	DS
12630004	970	25	28
12630005	853	16	17
12630013	1186	45	54
12630014	349	42	42
12630020	682	26	23
12630022	650	43	34
12630034	792	53	55

Table 7 ha-PA results for the adapted HIC revalidation runs

DS batch	Amount of ha-PA [ppm]		
	Cation exchange chromatography	Adapted HIC	DS
12650000	718	154	159
12650001	885	199	202
12650002	954	190	180

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Analytical procedure
Residual ha-PA is measured using an ELISA. The ha-PA ELISA is a solid phase, two-site enzyme immunoassay based on the direct sandwich technique in which two monoclonal antibodies are directed against different epitopes of the ha-PA molecule. During incubation, ha-PA contained in the test samples reacts with peroxidase-conjugated anti-ha-PA antibodies and anti-ha-PA antibodies bound to the well of a microtiter plate. A washing step removes unbound enzyme-labelled antibody. The bound conjugate is detected by reaction of the peroxidase with o-phenylendiamine dihydrochloride. The reaction is stopped by adding acid to give a colourimetric endpoint that is read spectrophotometrically. The content of ha-PA is calculated from a standard curve.

The LOQ of the analytical method is 0.5 ng/mL.

Summary
The ha-PA removal capacity of the tenecteplase purification process was sufficiently shown for materials from the manufacturing scale.

2.4 INSULIN

Insulin is used as growth factor in the cell culture media. In the tenecteplase purification process, it is removed mainly by the ELAC affinity chromatography. Theoretical clearance calculation based on worst-case assumptions in a fate assessment resulted in 797 µg insulin per maximum daily dose of 50 mg protein well above the PDE of 1.0 µg/dose.

The initial process validation studies (DS batches 22000 – 22002, refer to Section 3.2.S.2.5 Process Validation and / or Evaluation) and analytical testing of retains of in-process and DS samples from already released commercial manufacturing batches (Table 8) demonstrated that already after ELAC affinity chromatography the measured values are below the LOD or LOQ, respectively, in all investigated batches. Values provided for in-process samples and DS were calculated theoretically considering LOD or LOQ, sample dilution and protein concentration.

Insulin is considered as diabetic agent. The PDE was established as part of the PRI assessment to 1.0 µg/day. For a maximum dose of 50 mg, this correlates to a maximum allowable insulin concentration of 20 ng/mg tenecteplase. All measured values for ELAC affinity chromatography and DS are below the PDE. Based on the DS values shown in Table 8, the theoretical maximum insulin application per daily dose of 50 mg tenecteplase DS would be no more than 400 ng.

Since the removal capability of the tenecteplase purification process regarding insulin has been shown to be efficient and reproducible for all batches investigated, monitoring was discontinued.

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Table 8 Insulin removal during purification

DS batch	Amount of insulin [ng/mg]		
	Harvested cell culture fluid	ELAC affinity chromatography	DS
12630044	< 55 [a]	< 7 [a]	< 8 [a]
12630045	< 53 [a]	< 7 [a]	< 8 [a]
12630046	< 55 [a]	< 7 [a]	< 8 [a]

a value below LOQ, thus value calculated theoretically considering LOQ, sample dilution and protein concentration

Analytical procedure
Residual insulin is measured using an ELISA. The insulin ELISA is a solid phase, two-site enzyme immunoassay based on the direct sandwich technique in which two monoclonal antibodies are directed against different epitopes of the insulin molecule. During incubation, insulin contained in the test samples reacts with peroxidase-conjugated anti-insulin antibodies and anti-insulin antibodies bound to the well of a microtiter plate. A washing step removes unbound enzyme-labelled antibody. The bound conjugate is detected by reaction of the peroxidase with 3,3´,5,5´-tetramethylbenzidine. The reaction is stopped by adding acid to give a colourimetric endpoint that is read spectrophotometrically. The content of insulin is calculated from a standard curve.
The LOQ of the analytical method used for the measurement of retains of commercial manufacturing batches is 0.375 ng/mL.

Summary
The purification tenecteplase process reproducibly reduces insulin to acceptable levels below the LOQ, i.e., to a concentration below 8 ng/mg protein.
Thus, insulin removal capacity of the purification process was sufficient. Based on data presented here, insulin levels in DS are more than 2.5-fold below the TTC of 1.0 µg for a maximum daily dose of 50 mg tenecteplase.

2.5 POLOXAMER 188

The non-ionic surfactant poloxamer 188 is a component of the culture media as protectant for mechanical and sparge-related shear. In the tenecteplase purification process, it is removed mainly by the ELAC affinity chromatography. Theoretical clearance calculation based on worst-case assumptions in a fate assessment resulted in 81,126 µg poloxamer 188 per maximum daily dose of 50 mg protein well above the PDE of 16,500 µg/dose.

Retains of in-process and DS samples from already released commercial manufacturing batches were analysed to determine the removal of poloxamer 188. As shown in Table 9, already after ELAC affinity chromatography the poloxamer 188 concentration is at or below the LOQ in all investigated batches. Values provided for affinity chromatography and 10 kDa II UF / DF were calculated theoretically considering LOQ and protein concentration. DS was not analysed due to interfering polysorbate 20.

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The PDE was established as part of the PRI assessment to 16,500 µg/day. For a maximum dose of 50 mg, this correlates to a maximum allowable poloxamer 188 concentration of 330 µg/mg tenecteplase. All measured values are below the PDE. The theoretical maximum poloxamer 188 application per daily dose of 50 mg tenecteplase DS would be no more than 10 µg.

Since the removal capability of the tenecteplase purification process regarding poloxamer 188 has been shown to be efficient and reproducible for all batches investigated, monitoring was discontinued.

Table 9 Poloxamer 188 removal during purification

DS batch	Amount of poloxamer 188 [µg/mg]	
	ELAC affinity chromatography	10 kDa II UF / DF
12630044	≤ 0.5 [a]	≤ 0.2 [a]
12630045	≤ 0.5 [a]	≤ 0.2 [a]
12630046	≤ 0.5 [a]	≤ 0.2 [a]

a Value below LOQ, thus value calculated theoretically considering LOQ and protein concentration

Analytical procedure
Poloxamer 188 in the supernatant is detected using high performance liquid chromatography / electrospray ionization-tandem mass spectrometry (LC-ESI-MS/MS).
Poloxamer 188 is analysed in test samples after protein removal by acetonitrile precipitation. The peak areas of poloxamer 188 of each sample preparation are compared to the mean peak area of poloxamer 188 (2 µg/mL limit area) of the two consecutive preparations of spiked sample bracketing the sample to be tested.

The LOQ of the analytical method is 2 µg/mL.

Summary
The poloxamer 188 removal capacity of the purification process is sufficient and highly effective. Based on data presented here, poloxamer 188 levels in DS are at least 1,650-fold below the TTC of 16,500 µg for a maximum daily dose of 50 mg tenecteplase.

2.6 GENTAMICIN

Gentamicin is added to the cell culture media during the cell culture process as an antibiotic agent to minimise the risk of bacterial contamination of the media prior to sterile filtration. In the purification process, gentamicin is removed during chromatography and UF / DF steps. Theoretical clearance calculation based on worst-case assumptions in a fate assessment resulted in 0.17 ng gentamicin per maximum daily dose of 50 mg protein well below the TTC of 5.0 µg.

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Applying an ELISA, the initial process validation studies demonstrated that the commercial manufacturing process is capable to reproducibly remove gentamicin to levels below the LOD of no more than 14 ppm, corresponding to 14 ng/mg tenecteplase, after HIC (DS batches 22000 – 22002, refer to Section 3.2.S.2.5 Process Validation and / or Evaluation).

Furthermore, in a spiking study performed at small scale, the gentamicin removal capacity of the chromatography and the final 10 kDa UF / DF steps was assessed. As shown in Table 10, gentamicin was reduced by a total factor of at least $1.5 * 10^{10}$ (= 10.18 $\log_{10}$) at a spike addition at about 10 µg gentamicin per mL in the load materials of the evaluated unit operations.

Table 10 Evaluation of gentamicin removal capability in spike studies

Unit operation	Process intermediate	Volume [mL]	Gentamicin [µg/mL]	Reduction factor
ELAC affinity chromatography	Load	171	11.2	> 373
	Eluate	73	< 0.030 [a]	
Cation exchange chromatography	Load	99	11.0	> 367
	Eluate	62	< 0.030 [a]	
HIC	Load	53	12.1	> 403
	Eluate	126	< 0.030 [a]	
10 kDa II UF / DF	Load	1229	10.7	282
	Retentate	63	0.038	
			Total reduction factor	*> 1.5 * 10^{10} (= 10.18 $\log_{10}$)*

a Value below LOQ of 0.030 µg/mL of the gentamicin ELISA

In addition, analytical testing of in-process and DS samples from already released commercial manufacturing batches by high performance liquid chromatography with fluorescence detection (HPLC-FLD) was performed. As shown in Table 11, already after ELAC affinity chromatography the gentamicin concentration is below the LOQ in all investigated batches and values were calculated theoretically considering LOQ, sample dilution and protein concentration.

The TTC was established as part of the PRI assessment to 5.0 µg/day. For a maximum dose of 50 mg, this correlates to a maximum allowable gentamicin concentration of 0.1 µg/mg tenecteplase. After the tenecteplase purification process, the measured values are above the TTC. Based on the DS values shown in Table 11, the maximum gentamicin application per daily dose of 50 mg tenecteplase DS would be no more than 10.0 µg. However, all values obtained were below LOQ. Based on the visual evaluation of the chromatographic profiles, the signal of the un-spiked DS samples is at the level of the blank samples compared to the

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DS samples spiked with 1 µg gentamicin per mL. Thus, the gentamicin concentration in the DS samples is far below 1 µg/mL and thus far below the TTC of 5.0 µg.

Since the removal capability of the purification process regarding gentamicin has been shown to be efficient and reproducible for all batches investigated, monitoring was discontinued.

Table 11 Gentamicin removal during purification

DS batch	Amount of gentamicin [µg/mg]		
	Harvested cell culture fluid	ELAC affinity chromatography	DS
12630044	199.96	< 0.26 [a]	< 0.20 [a]
12630045	183.50	< 0.26 [a]	< 0.20 [a]
12630046	190.86	< 0.27 [a]	< 0.20 [a]

a value below LOQ, thus value calculated theoretically considering LOQ, sample dilution and protein concentration

Analytical procedure
Gentamicin is detected using HPLC-FLD. Gentamicin is analysed in test samples after addition of internal standard DL-2-aminobutyric acid and labelling of gentamicin amino groups, which is facilitated by reaction with 6-aminoquinolyl-N-hydroxysuccinimidyl carbamate. After derivatisation, the samples are analysed (diluted or undiluted) and detected using excitation wavelength of 250 nm and emission wavelength of 395 nm. For harvested cell culture fluid, the numerical analytical result is reported. The content of gentamicin is calculated from a standard curve. For the affinity chromatography and DS samples a limit test was applied. The peak areas of gentamicin of each sample preparation are compared to the mean peak area of the same samples spiked at 1 µg/mL (limit area).

The LOQ of the analytical method is 1.0 µg/mL.

Summary
The gentamicin removal capacity of the purification process is sufficient and highly effective. Based on data presented here, gentamicin levels in DS are below the TTC of 5.0 µg for a maximum daily dose of 50 mg tenecteplase.

2.7 METHOTREXATE

MTX is added during the inoculum steps as selective agent to the cell culture medium. In the subsequent cell cultivation stages (80 L to 2,000 L expansion bioreactors and 12,000 L production bioreactor) MTX is not added to the cell culture media and therefore significantly diluted until harvest stage. In the purification process, MTX is removed during chromatography and UF / DF steps. Theoretical clearance calculation based on worst-case

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assumptions in a fate assessment resulted in 0.008 pg MTX per maximum daily dose of 50 mg protein well below the TTC of 1.5 µg.

Retains of in-process and DS samples from already released commercial manufacturing batches were analysed to determine the removal of MTX. As shown in Table 12, already after ELAC affinity chromatography all results were below the LOQ in all investigated batches. Values were calculated theoretically considering LOQ and protein concentration. Due to matrix interference the MTX concentration in the harvested cell culture fluid could not be determined.

The TTC was established as part of the PRI assessment to 1.5 µg/day due to the genotoxic potential of MTX. It is considered a worst case TTC, because it is based on lifetime exposure. For a maximum dose of 50 mg, this correlates to a maximum allowable MTX concentration of 30 ng/mg tenecteplase. All measured values are below the TTC. The theoretical maximum MTX application per daily dose of 50 mg tenecteplase DS would be less than 0.10 µg.

Since the removal capability of the purification process regarding MTX has been shown to be efficient and reproducible for all batches investigated, monitoring was discontinued.

Table 12 MTX removal during purification

DS batch	Amount of methotrexate [ng/mg]	
	ELAC affinity chromatography	DS
12630044	< 2.6 [a]	< 2.0 [a]
12630045	< 2.6 [a]	< 2.0 [a]
12630046	< 2.7 [a]	< 2.0 [a]

a value below LOQ, thus value calculated theoretically considering LOQ and protein concentration

Analytical procedure
The analytical method was developed as a limit test for monitoring of MTX in aqueous protein matrix samples using LC-ESI-MS/MS.
Protein contained in the samples is precipitated by the addition of methanol. MTX loss during this precipitation process is monitored by an internal deuterated standard, differing in mass from the MTX used for production. During MS / MS fragmentation a very stable daughter ion of MTX is generated by the loss of glutamic acid. The internal deuterated MTX standard is spiked at the reporting limit concentration (10 ng/mL). MTX results are calculated by the ratio of the peak area of the corresponding sample peak of the daughter ion in the chromatogram and the corresponding internal deuterated standard daughter ion peak.

The LOQ of the analytical method is 10 ng/mL.

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Summary
The MTX removal capacity of the tenecteplase purification process is sufficient and highly effective. Based on data presented here, MTX levels in DS are more than 15-fold below the TTC of 1.5 µg for a maximum daily dose of 50 mg tenecteplase.

2.8 PHENOL RED

Phenol red is a component of the culture media in the inoculum steps. In the tenecteplase purification process, phenol red is removed during chromatography and UF / DF steps. Theoretical clearance calculation based on worst-case assumptions in a fate assessment resulted in 0.046 pg phenol red per maximum daily dose of 50 mg protein well below the TTC of 1.5 µg/dose.

Retains of in-process and DS samples from already released commercial manufacturing batches were analysed to determine the removal of phenol red. As shown in Table 13, already after ELAC affinity chromatography the phenol red concentration is below the LOQ in all investigated batches and values were calculated theoretically considering LOQ and protein concentration.

The TTC was established as part of the PRI assessment to 1.5 µg/day due to the genotoxic potential of phenol red. It is considered a worst case TTC because it is based on lifetime exposure. For a maximum dose of 50 mg, this correlates to a maximum allowable phenol red concentration of 30 ng/mg tenecteplase. All measured values are below the TTC. The theoretical maximum phenol red application per daily dose of 50 mg tenecteplase DS would be less than 5.0 ng.

Since the removal capability of the tenecteplase purification process regarding phenol red has been shown to be efficient and reproducible for all batches investigated, monitoring was discontinued.

Table 13 Phenol red removal during purification

DS batch	Amount of phenol red [ng/mg]		
	Harvested cell culture fluid	ELAC affinity chromatography	DS
12630044	15.7	< 0.1 [a]	< 0.1 [a]
12630045	14.8	< 0.1 [a]	< 0.1 [a]
12630046	13.7	< 0.1 [a]	< 0.1 [a]

a value below LOQ, thus value calculated theoretically considering LOQ and protein concentration

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Analytical procedure
Phenol red is quantified using reversed phase high performance liquid chromatography (RP-HPLC) with ultraviolet / visible detection at 557 nm. Prior to analysis, the protein from the sample matrix is precipitated at 70°C using methanol.
The LOQ of the analytical method is 0.5 ng/mL.

Summary
The phenol red removal capacity of the purification process is sufficient and highly effective. Based on data presented here, phenol red levels in DS are more than 300-fold below the TTC of 1.5 µg for a maximum daily dose of 50 mg tenecteplase.

2.9 SIMETHICONE

Simethicone is added during the cell culture process at the expansion bioreactor and production bioreactor stages as an anti-foaming agent. Due to the hydrophobic nature of the molecule, it is removed from the product during the purification process mainly by the ELAC affinity chromatography. Theoretical clearance calculation based on worst-case assumptions in a fate assessment resulted in 1,065 µg simethicone per maximum daily dose of 50 mg protein well below the PDE of 2,500 µg/dose.

Retains of in-process and DS samples from already released commercial manufacturing batches were analysed to determine the removal of simethicone. As shown in Table 14, already after ELAC affinity chromatography the simethicone concentration is at or below the LOQ in all investigated batches and values were calculated theoretically considering LOQ, sample dilution and protein concentration.

The PDE was established as part of the PRI assessment to 2,500 µg/day. For a maximum dose of 50 mg, this correlates to a maximum allowable simethicone concentration of 50 µg/mg tenecteplase. All measured values after ELAC affinity chromatography are below the PDE. The maximum simethicone application per daily dose of 50 mg tenecteplase DS would be 5.0 µg.

Since the removal capability of the tenecteplase purification process regarding simethicone has been shown to be efficient and reproducible for all batches investigated, monitoring was discontinued.

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Table 14 Simethicone removal during purification

DS batch	Amount of simethicone [µg/mg]		
	Harvested cell culture fluid	ELAC affinity chromatography	DS
12630044	90.9	< 0.1 [a]	< 0.1 [a]
12630045	87.4	< 0.1 [a]	< 0.1 [a]
12630046	100.0	< 0.1 [a]	0.1

a value below LOQ, thus value calculated theoretically considering LOQ, sample dilution and protein concentration

Analytical procedure
Simethicone is quantified using inductively coupled plasma optical emission spectrometry (ICP-OES) detection of elemental silicon (Si). The sample is diluted with water (dilution factor 10) and analysed by ICP-OES. Quantification of residual simethicone in samples is performed versus a calibration curve for elemental Si and calculation to polydimethylsiloxane (PDMS) using the theoretical Si content in PDMS (37.87 % according to the sum formula $((CH3)_2SiO)n$).

The LOQ of the analytical method is 0.026 µg/mL PDMS.

Summary
The simethicone removal capacity of the tenecteplase purification process is sufficient and highly effective. Based on data presented here, simethicone levels in DS are 500-fold below the TTC of 2,500 µg for a maximum daily dose of 50 mg tenecteplase.

2.10 PROPYLENE GLYCOL

Propylene glycol is a component of the elution buffer used during the HIC step and is removed by the final 10 kDa UF / DF step. Theoretical clearance calculation based on worst-case assumptions in a fate assessment resulted in 196 µg propylene glycol per maximum daily dose of 50 mg protein well below the PDE of 125,000 µg/dose.

The initial process validation studies (DS batches 22000 – 22002, refer to Section 3.2.S.2.5 Process Validation and / or Evaluation) demonstrated that the final 10 kDa UF / DF is capable to reproducibly remove propylene glycol to levels below 1 µg/mg. The maximum level was 0.955 µg/mg tenecteplase.

The PDE was established as part of the PRI assessment to 125,000 µg/day. For a maximum dose of 50 mg, this correlates to a maximum allowable propylene glycol concentration of 2,500 µg/mg tenecteplase. All measured values after the UF / DF step are below the PDE for propylene glycol. The maximum propylene glycol application per daily dose of 50 mg tenecteplase DS would be 48 µg.

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Since the removal capability of the purification process regarding propylene glycol has been shown to be efficient and reproducible for all batches investigated, monitoring was discontinued.

Analytical procedure
Propylene glycol is quantified using gas chromatography with flame ionization detection. Before analysis, the sample is diluted in water by factor 10. Quantification is performed by normalisation procedure, i.e., determination of the area of propylene glycol as a percentage of the total area of all detected peaks. The disregard limit for others than the propylene glycol peak is equal to 0.025 mg/mL.

Summary
The propylene glycol removal capacity of the tenecteplase purification process is sufficient and highly effective. Based on data provided in the process validation (refer to Section 3.2.S.2.5 Process Validation and / or Evaluation), propylene glycol levels in DS are 2,604-fold below the PDE of 125,000 µg for a maximum daily dose of 50 mg tenecteplase.

3. PRODUCT-RELATED IMPURITIES

Product-related impurities are molecular variants of the active molecule that may arise during manufacture of the DS (Table 15).

The tenecteplase purification process was developed to control and remove HMW and LMW species. The robust removal of product-related impurities during DS manufacture was demonstrated via initial process validation (DS batches 22000 – 22002, refer to Section 3.2.S.2.5 Process Validation and / or Evaluation) and testing of in-process and DS samples of commercial manufacturing batches and adapted HIC revalidation batches provided in Chapters 3.1 and 3.2. During DS storage, constant levels of product-related impurities were demonstrated (refer to Section 3.2.S.7.3 Stability data – Accelerated and Long-Term Stability Studies).

Table 15 Product-related impurities

Process-related impurity	Source	Method of control
HMW species	Process conditions during DS manufacturing	In-process testing during PPQ, revalidation and in selected commercial manufacturing batches; DS specification (indirect by determination of monomers)
LMW species	Process conditions during DS manufacturing	In-process testing during PPQ, revalidation and in selected commercial manufacturing batches; DS specification

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3.1 HIGH MOLECULAR WEIGHT SPECIES

During process validation, HMW species were indirectly monitored by the determination of monomers in DS applying high performance size exclusion chromatography (HPSEC). Data from process validation and revalidation studies have shown that the tenecteplase purification process reproducibly yields a monomer content of 97 % to 100 % (refer to Sections 3.2.S.2.5 Process Validation and / or Evaluation and 3.2.S.2.6 Manufacturing Process Development). In addition, HMW species were indirectly controlled via sodium dodecyl sulphate polyacrylamide gel electrophoresis (SDS-PAGE). All batches complied with tenecteplase reference material.

In addition, analytical testing of in-process samples from already released commercial manufacturing batches by HPSEC was performed. Data were compared to DS release data. As shown in Table 16, the purification capability of the tenecteplase purification process regarding HMW species has been shown to be efficient and reproducible for all investigated batches. This also applies to the adapted HIC revalidation batches shown in Table 17. Thus, monitoring of HMW species in in-process samples was discontinued. HMW species are indirectly controlled as part of the DS release testing of the monomer content.

Table 16 HMW species content during purification

DS batch	Content of HMW species [%]				
	ELAC affinity chromatography	Acid treatment	Cation exchange chromatography	HIC	DS
12630004	5.9	2.9	1.4	0.2	0.6
12630005	6.0	2.8	1.2	0.2	0.6
12630013	6.1	2.5	1.2	0.2	0.5
12630014	5.1	2.2	1.2	0.2	0.5
12630020	4.5	2.1	1.0	0.2	0.4
12630022	5.3	2.4	1.0	0.2	0.6
12630034	5.9	2.5	1.0	0.1	0.6

Table 17 HMW species content in adapted HIC revalidation batches

DS batch	Content of HMW species [%]	
	Adapted HIC	DS
12650000	0.2	0.4
12650001	0.2	0.5
12650002	0.3	0.4

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Analytical procedure
The analytical procedure for HPSEC is the same as used for release testing of tenecteplase drug substance. For a description, refer to Section 3.2.S.4.2 Analytical Procedure.

Summary
The capacity of the tenecteplase purification process to maintain a high level of monomer and thus a low level of HMW species was demonstrated. Furthermore, the purification process is capable of producing a product of very high purity with respect to the monomer content of tenecteplase.

3.2 LOW MOLECULAR WEIGHT SPECIES

During process validation, LMW species were indirectly monitored by the determination of monomers in DS applying HPSEC. Data from process validation and revalidation studies have shown that the tenecteplase purification process for tenecteplase reproducibly yields a monomer content of 97 % to 100 % (refer to Section 3.2.S.2.5 Process Validation and / or Evaluation and 3.2.S.2.6 Manufacturing Process Development). In addition, LMW species were indirectly controlled via SDS-PAGE. All batches complied with tenecteplase reference material.

In addition, analytical testing of in-process and DS samples from already released commercial manufacturing batches by capillary gel electrophoresis (CGE) under reduced conditions was performed. As shown in Table 18, the purification capability of the tenecteplase purification process regarding LMW species monitored by the fragment region 1 has been shown to be efficient and reproducible for all investigated batches. This also applies to the adapted HIC revalidation batches shown in Table 19.

A low content of fragment region 1 is maintained throughout the purification process. Thus, monitoring of LMW species in in-process samples was discontinued.

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Table 18 LMW species content during purification

DS batch	Content of LMW species assessed by fragment region 1 content [%]				
	ELAC affinity chromatography	Acid treatment	Cation exchange chromatography	HIC	DS
12630004	0.5	0.5	0.4	0.3	0.4
12630005	0.4	0.4	0.3	0.2	0.4
12630013	0.6	0.5	0.4	0.3	0.4
12630014	0.6	0.5	0.4	0.3	0.4
12630020	0.5	0.4	0.3	0.4	0.3
12630022	0.5	0.4	0.3	0.2	0.4
12630034	0.6	0.5	0.3	0.2	0.3

Table 19 LMW species content in adapted HIC revalidation batches

DS batch	Content of LMW species assessed by fragment region 1 content [%]	
	Adapted HIC	DS
12650000	0.5	0.6
12650001	0.5	0.5
12650002	0.5	0.6

Analytical procedure
For the data shown in Table 18, the analytical procedure for CGE reduced is comparable to the method used for release testing of tenecteplase DS except for the sample preparation and that the peak consisting of the Arg27 clip (amino acids 1 to 27; for details refer to Section 3.2.S.3.1 Elucidation of Structure) is part of the peak integration for fragment region 1. For the data shown in Table 18 the peak area consisting of Arg27 clip was subtracted from the total fragment region 1 area. Data for the adapted HIC revalidation batches shown in Table 19 were generated with the release method valid at time of testing. For a description of the release method, refer to Section 3.2.S.4.2 Analytical Procedure.

Summary
The capacity of the tenecteplase purification process to maintain a sufficiently low level of fragment region 1 and thus a low level of LMW species was demonstrated.

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4. CONCLUSION

Potential impurities in tenecteplase DS manufacture have been classified as PRIs and product-related impurities.

The data obtained indicate that the PRIs investigated are efficiently and reproducibly removed from DS during the tenecteplase purification process.

Product-related impurities are efficiently controlled by the tenecteplase purification process. The results presented show a high purity of DS with regards to the amount of intact, monomeric tenecteplase that is maintained throughout the purification process until the DS level. No new product-related impurities are introduced during the manufacture of tenecteplase DP.